Jayakrishnan R

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.35
Saraswathi Vidyalaya Vattiyoorkavu Trivandrum	Class XII (Central Board of Secondary Education)	2019	95.00%
Saraswathi Vidyalaya Vattiyoorkavu Trivandrum	Class X (Central Board of Secondary Education)	2017	10.00%

PROJECTS

VOF-DPM Coupling and Evaporating Spray Modelling in a Jet-in-Crossflow

- Created the geometry for the jet in cross flow problem and generated the mesh in Fluent meshing .
- Validating the **VOF-to-DPM** transition methodology in ANSYS Fluent and evaluating spray characteristics - including **Sauter Mean Diameter , droplet size distribution,** and **jet penetration** against the reference study.
- Injecting droplets as DPM particles and investigating post-transition droplet dynamics, including velocity, trajectory, secondary breakup, droplet size evolution, and spatial distribution of the dispersed spray due to the crossflow.
- Extending the DPM framework to model evaporation of the droplets and analyzing its effects on droplet size, temperature, lifetime, and overall spray evolution.